

Find-Fast Answer Map

- **L1** data types, data life cycle, governance, FEAT, encoding, integration, "why finance data is hard"
- **L2** supervised/unsupervised/RL, logistic/tree/DL, confusion matrix, precision vs recall, ROC/AUC, overfitting.
- **L3** risk-return, CAPM, beta, factor investing, Sharpe/Sortino/Max DD, backtesting traps.
- **L4** NLP, BoW, embeddings, sentiment, Dow Theory, MA/MACD/RSI/Bollinger Bands.
- **L5** AML/CFT, KYC/CDD/EDD/SAR, FATF, K-means/DBSCAN/anomaly detection, graph 3Cs.
- **L6** agentic AI, Ollama, RAG, MCP, LangChain, LangGraph, workflow design.
- **Exam tactic** first match the keyword in the question to the lecture bucket above; then use the direct answer lines below.

L1 Intro to AI in Finance

- **Where AI is used** credit/loan approval, customer segmentation, prospecting, fraud/AML, chatbots, robo-advisory, risk management, trading, news monitoring, compliance reporting.
 - **Why finance is special** finance affects livelihoods and systemic stability; markets adapt; human behavior is noisy; data is messy; experiments are hard; regulation matters more than in many other domains.
 - **Finance data types structured numeric** prices, volume, rates, revenue, assets; **ordinal** ratings AAA->BB, analyst buy/hold/sell; **nominal** sector/industry labels; **unstructured** news, filings, audio, images, social posts.
 - **Alternative data** web traffic, card spending, satellite images, geolocation, weather, app usage, parking-lot activity; used to get an informational edge beyond standard statements/prices.
 - **Data sources** Bloomberg/Refinitiv, exchanges, SEC EDGAR/SGXNet, IMF/World Bank/FED, internal transaction systems, web/API sources.
 - **Transformation** aggregate daily->weekly/monthly; normalize with **min-max**; standardize with **z-score**; smooth, encode, clean, and create features so models can learn stable patterns.
 - **Integration issues** multiple sources may disagree; values can be duplicated, stale, missing, or inconsistent; poor integration creates garbage-in-garbage-out errors.
 - **Encoding** models need numbers; **one-hot encoding** turns categories into binary flags and avoids fake ordering between categories.
 - **Data governance** objective = data quality + privacy/security + standardization + accountability + business value; the point is not only to store data but to make it reliable and defensible.
 - **Data life cycle memory Plan -> Design -> Create/Obtain -> Store/Maintain -> Use -> Enhance.** If the exam asks for steps, keep this exact order.
 - **Plan** ■■■ define business purpose, what decision the data supports, who owns it, which fields are needed, legal basis/consent, retention period, success metric, and access rights.
 - **Design** ■■■ choose schema/table structure, labels, data format, update frequency, validation checks, privacy controls, and how downstream teams/models will consume the data.
 - **Create/Obtain** ■■■ collect from customers, internal systems, vendors, public filings, or exchanges; verify licensing, provenance, consent, completeness, and initial quality.
 - **Store/Maintain** ■■■■■ keep in database/data lake/cloud; add backups, encryption, access control, versioning, lineage, cleaning, deduplication, and regular updates.
 - **Use** ■■■ apply the data to reporting, model training, segmentation, pricing, AML monitoring, trading, or decision support, but only for the approved business purpose.
 - **Enhance** ■■■ enrich with external sources, engineer new features, relabel/repair errors, anonymize where needed, and learn from user feedback to improve quality and usefulness.
 - **MAS FEAT Fairness** = unbiased and consistent outcomes; **Ethics** = use AI responsibly; **Accountability** = the institution owns the decision and cannot blame the model; **Transparency** = explainable, documented, auditable logic.
 - **Q** Which FEAT principle says the firm cannot blame the algorithm? **Ans** Accountability.
 - **Q** Which data type fits analyst recommendations or credit ratings? **Ans** Ordinal data.
 - **Q** Why do one-hot encoding? **Ans** To avoid false ordinal relationships between categories.
- ## L2 ML in Finance Operations
- **ML types supervised** uses labeled outcomes; **unsupervised** finds hidden patterns without labels; **reinforcement learning** learns actions from reward/penalty feedback.
 - **Typical supervised tasks classification** predicts a class (default / non-default, fraud / not fraud); **regression** predicts a number (return, loss, price, revenue).
 - **Workflow** define objective -> gather labeled data -> select features -> split train/validation/test -> train -> evaluate on unseen data -> tune threshold/model -> deploy -> monitor drift and errors.
 - **Data splits resubstitution** train=test (weak, only if data is tiny); **hold-out** often 80/20; **cross-validation** rotates folds for more stable evaluation; **bootstrap** samples with replacement.
 - **Why validation matters** finance is non-stationary and noisy, so high in-sample accuracy can be fake comfort;

only out-of-sample performance shows whether the model generalizes.

- **Common models** logistic regression, naive Bayes, decision trees, SVM, perceptron/deep learning; no single model is always best.
- **Logistic regression** interpretable baseline for binary outcomes; outputs probability and odds $p/(1-p)$; often a threshold such as 0.5 maps probability to class.
- **Decision tree** easy to explain and audit; works with numeric + categorical variables; downside = unstable and can overfit if the tree becomes too complex.
- **Naive Bayes** fast probabilistic benchmark based on Bayes rule; assumes features are independent, which is often unrealistic but still useful as a simple baseline.
- **Deep learning** powerful for complex nonlinear patterns and unstructured data, but is compute-heavy, less explainable, and can overfit.
- **Confusion matrix TP** predicted positive and actually positive; **TN** predicted negative and actually negative; **FP** false alarm; **FN** missed bad case.
- **Accuracy** = overall correctness, but it can look great when classes are imbalanced. Example: if 95% of loans are good, always predicting "good" gives high accuracy but poor risk control.
- **Precision** $TP/(TP+FP)$ = among the cases you flagged positive, how many were truly positive? Use when false alarms are costly.
- **Recall / Sensitivity** $TP/(TP+FN)$ = among the truly positive cases, how many did you catch? Use when misses are costly.
- **Threshold trade-off** lower threshold -> catch more positives -> higher recall but also more FP; higher threshold -> fewer FP but more missed positives.
- **Default example** if "positive" means default, then **FP** = reject a customer who would have repaid; **FN** = approve a borrower who later defaults. Banks usually fear FN more.
- **ROC/AUC** ROC plots TPR against FPR across thresholds; **AUC** summarizes discrimination skill, with 1.0 = perfect and 0.5 = no better than random.
- **Overfitting** model memorizes training noise/details and performs badly on new data; **underfitting** model is too simple and misses real structure.
- **Finance overfitting traps** small samples, regime changes, repeated backtests, too many features, data snooping, and optimizing to past noise.
- **Metric logic** catching fraud/default -> recall often matters more; prospecting/marketing -> precision often matters more; always match metric to business cost of error.
- **Q** Conservative bank wants to catch defaulters. Best metric? **Ans** Recall / sensitivity.
- **Q** Marketing wants fewer false alarms on leads. Best metric? **Ans** Precision.
- **Q** 99% training accuracy but weak test accuracy indicates? **Ans** Overfitting.

L3 Investment Management

- **Risk-return idea** most investors are risk-averse: they want more return only if compensated for taking more risk; that extra expected compensation is the **risk premium**.
- **Return** = (ending value - invested amount) / invested amount; if dividends/coupons exist, include them in total return.
- **Expected return** is the probability-weighted average of possible returns, not just one scenario.
- **Risk types stand-alone risk** = total risk of one asset; **portfolio risk** depends on how assets move together, so diversification can reduce unsystematic risk.
- **Standard deviation** measures volatility / dispersion of returns; higher standard deviation = more uncertainty.
- **CAPM** $r_i = r_{RF} + (r_{M} - r_{RF})\beta_i$. Read it as: required return = risk-free rate + beta-adjusted market risk premium.
- **CAPM variables** r_{RF} = risk-free rate; $r_{M} - r_{RF}$ = market risk premium; β_i = how sensitive the stock is to market moves.
- **Beta meaning** $\beta=1$ moves like the market; $\beta>1$ is more aggressive/riskier; $\beta<1$ is more defensive; beta is about systematic market risk, not firm-specific noise.
- **Why only market risk in CAPM?** because diversification can remove firm-specific risk, so investors are mainly compensated for non-diversifiable market risk.
- **Factor pricing** extends CAPM beyond one factor: expected return can depend on market, size, value, momentum, profitability, investment, and other signals.
- **Factor examples** P/E, P/B, EV/EBITDA, EV/Sales, ROE, sentiment, macro factors, alternative data; AI/ML helps discover nonlinear or hidden factors.
- **Sharpe ratio** $(R_p - R_f)/\sigma_p$ = excess return per unit of total volatility; good for overall risk-adjusted performance.
- **Sortino ratio** $(R_p - R_f)/\sigma_d$ = excess return per unit of downside risk only; useful when upside volatility should not be punished.
- **Max drawdown** worst peak-to-trough loss; important because a deep drawdown is psychologically and financially hard to recover from.
- **Backtesting** simulate the strategy on historical data to see returns, stability, and drawdowns before live deployment.
- **Backtest traps** survivorship bias, short sample window, choosing the year that works, ignoring costs/slippage, data snooping, and worshipping one metric only.
- **Bagging vs boosting** bagging trains many weak learners in parallel then votes; **boosting** trains sequentially and focuses on prior errors.
- **Q** $\beta > 1$ implies what? **Ans** The stock is more

sensitive/riskier than the market.

- **Q** Which metric penalizes only downside volatility? **Ans** Sortino ratio.

- **Q** Great backtest but bad future live result usually means? **Ans** Overfitting, regime change, or data snooping.

L4 NLP + Algorithmic Trading

- **NLP purpose** convert unstructured text into signals that models can use for classification, sentiment, tagging, search, or decision support.
- **Text sources** news, analyst reports, filings, transcripts, contracts, emails, support chats, social media, policy statements, customer feedback.
- **BoW** Bag-of-Words counts terms in documents and builds a term-document matrix; it is simple and fast but loses word order and context.
- **Tokenization** break text into tokens/subwords/roots so a computer can process it; this helps standardize variants of the same word.
- **Embeddings** represent words or texts as vectors that preserve contextual similarity; unlike BoW, they keep semantic relationships better.
- **LLMs** large transformer-based models trained on massive corpora; useful for summarization, classification, Q&A, drafting, and workflow support.
- **Finance NLP uses** sentiment on news, entity/event extraction, compliance screening, KYC name matching, chatbot support, product matching, risk summarization.
- **Technical analysis** studies price/volume charts for pattern, trend, and timing; AI can augment TA by automating signal extraction.
- **Dow Theory** market discounts everything; primary trend matters most; trends have accumulation -> public participation -> distribution phases.
- **Moving averages** smooth noise and help identify trend direction/support/resistance; crossover rules often generate simple buy/sell signals.
- **MACD** 12-EMA minus 26-EMA; signal line = 9-EMA of MACD; crossovers or divergences hint at momentum changes.
- **RSI** momentum oscillator; below 30 often = oversold; above 80 often = overbought.
- **Bollinger Bands** SMA with upper/lower volatility bands; squeeze/tightening can warn of an upcoming breakout.
- **TA pros/cons** pros = intuitive and easy to apply; cons = subjective, not strongly grounded in theory, and can miss surprise fundamentals.
- **Q** Biggest weakness of BoW? **Ans** It loses word order and deeper context.
- **Q** RSI > 80 means? **Ans** Overbought. RSI < 30 means oversold.
- **Q** MACD crossing above signal line often suggests? **Ans** A possible bullish/buy signal.

L5 Regulatory Compliance + Unsupervised Learning

- **Compliance split regulatory compliance** = obey laws/rules/sanctions/Base; **financial crime compliance** = AML/CFT/fraud detection and reporting.
 - **Money laundering stages Placement** put dirty money into the system; **Layering** move it through complex transfers to hide origin; **Integration** reintroduce it as apparently legitimate wealth.
 - **FATF** global AML standard setter; the FATF 40 Recommendations emphasize risk-based controls, KYC, suspicious reporting, cooperation, beneficial ownership, and sanctions.
 - **KYC/CDD/EDD/SAR** KYC verify customer identity; **CDD** understand purpose, source of funds, expected activity, and risk; **EDD** deeper checks for high-risk clients; **SAR** suspicious activity report filed when behavior looks abnormal.
 - **Red flags** just-below-threshold transfers, rapid movement through many accounts, shell companies, inconsistent explanations, refusal to provide documents, high-risk jurisdictions, sanctions exposure.
 - **Why AI matters** institutions face huge alert volumes; AI helps prioritize suspicious activity, reduce false positives, trace patterns, and monitor adverse news.
 - **Unsupervised learning** finds clusters, structure, or anomalies without labeled targets; useful when suspicious behavior is rare or not fully labeled.
 - **K-means** choose k, assign points to nearest centroid, recompute centroids, and repeat; best for roughly compact/spherical clusters.
 - **Distance measures** Euclidean, Manhattan, Minkowski; standardize numeric features first so one variable does not dominate only because of scale.
 - **Anomaly detection** flags points that do not fit the normal pattern; perfect for fraud, AML, unusual trading, or operational risk monitoring.
 - **DBSCAN** density-based clustering with eps + minPts; good for irregular shapes, noisy data, and outliers, so it often fits suspicious-activity patterns better than K-means.
 - **Graphs/networks** represent entities as nodes and relationships as links; useful for KYC ownership webs, contagion, payment networks, and exposure chains.
 - **3 Cs centrality** who matters most; **clusterness** who groups together; **connectedness** how risk/influence spreads across the network.
 - **Q** Correct laundering order? **Ans** Placement -> Layering -> Integration.
 - **Q** Best method for irregular suspicious clusters plus outliers? **Ans** DBSCAN.
 - **Q** What is the operational pain point in AML systems? **Ans** Too many false positives.
- ## L6 Agentic AI in Finance
- **Agentic AI** autonomous, goal-directed systems that

perceive, reason, use tools, act, and adapt; it is about getting work done, not only generating text.

- **Agentic AI vs GenAI** **GenAI** mainly generates or summarizes; **agentic AI** plans steps, calls tools/APIs, uses memory, and completes multi-step workflows.
- **How it works** perception/input -> reasoning/planning -> action/tool use -> feedback -> adjustment toward goal.
- **Ollama** local runtime for open-source LLMs; good for privacy, cost control, offline use, latency, and regulatory comfort.
- **RAG** Retrieval-Augmented Generation: pull relevant knowledge from internal/external documents into the prompt so answers are grounded in actual sources.
- **MCP** Model Context Protocol: standardized bridge that lets the model connect to live tools, APIs, databases, spreadsheets, and services.
- **RAG vs MCP** if the task is **answer from documents**, choose **RAG**; if the task is **query or act on live systems/tools**, choose **MCP**. Easy exam memory: **RAG = knowledge, MCP = action**.
- **Framework roles** **LlamaIndex** helps RAG/indexing; **LangChain** general LLM app framework; **LangGraph** strong for multi-agent orchestration/workflow graphs.
- **Enterprise concerns** security, confidentiality, scaling, structured JSON outputs, robustness, customization, memory, and integration with legacy systems.
- **Finance use cases** client support, portfolio/risk reporting, FX hedging agents, trading assistants, compliance review, document summarization, internal knowledge search.
- **Design checklist** define goal -> map workflow -> decide what context is needed -> choose model -> choose tools -> set memory/guardrails -> add human oversight.
- **Q** "Summarize the risk section from a prospectus" needs what? **Ans** RAG.
- **Q** "Query my spending categories from a live database" needs what? **Ans** MCP.
- **Q** Which framework is strongest for multi-agent orchestration? **Ans** LangGraph.

If You Panic in the Exam, Use These Direct Answer Cues

- **Need labels?** supervised. **Need patterns without labels?** unsupervised. **Need reward-based action learning?** reinforcement learning.
- **Want fewer missed fraud/default cases?** choose recall. **Want fewer false alarms?** choose precision.
- **Single-factor pricing model?** CAPM. **Multi-factor extension?** factor model.
- **Total-risk performance metric?** Sharpe. **Downside-only metric?** Sortino.
- **Text counts only?** BoW. **Context-aware text vectors?** embeddings.
- **Known k, compact clusters?** K-means. **Irregular clusters + noise/outliers?** DBSCAN.
- **Document grounding?** RAG. **Live tool/database connection?** MCP.
- **RSI <30** oversold. **RSI >80** overbought. **β >1** more market-sensitive/riskier than market.

Formula Box

- **Min-max** $x' = (x - \min) / (\max - \min)$.
- **z-score** $z = (x - \mu) / \sigma$.
- **Accuracy** $(TP + TN) / (TP + TN + FP + FN)$.
- **Precision** $TP / (TP + FP)$.
- **Recall** $TP / (TP + FN)$.
- **CAPM** $r_i = r_{RF} + (r_M - r_{RF})\beta_i$.
- **Sharpe** $(R_p - R_f) / \sigma_p$.
- **Sortino** $(R_p - R_f) / \sigma_d$.
- **MACD** $12\text{-EMA} - 26\text{-EMA}$; signal = $9\text{-EMA}(\text{MACD})$.
- **RSI** $100 - 100 / (1 + n_{up} / n_{down})$.

Rapid Answer Bank

- **Q** Exact data life-cycle order? **Ans** Plan -> Design -> Create/Obtain -> Store/Maintain -> Use -> Enhance.
- **Q** Why is finance difficult for AI? **Ans** adaptive markets, human behavior, noisy data, regulation, and systemic risk.
- **Q** Which metric becomes misleading with class imbalance? **Ans** Accuracy.
- **Q** What does a lower classification threshold do? **Ans** raises recall and false positives.
- **Q** What does beta measure? **Ans** sensitivity of a stock to market movements.
- **Q** CAPM compensation is for which risk? **Ans** systematic / market risk.
- **Q** Why can factor premiums disappear? **Ans** crowding, regime change, or arbitrage reducing mispricing.
- **Q** BoW versus embeddings in one line? **Ans** counts only vs context-aware vectors.
- **Q** What does RSI below 30 usually mean? **Ans** oversold.
- **Q** KYC vs CDD in one line? **Ans** identity verification vs broader understanding of customer purpose/source/risk.
- **Q** Why is DBSCAN useful in AML? **Ans** it handles irregular suspicious clusters, density patterns, and outliers.
- **Q** What makes DBSCAN different from K-means? **Ans** DBSCAN is density-based; K-means uses distance to centroid and fixed k.
- **Q** Why standardize before clustering? **Ans** to equalize scales so big-number features do not dominate distances.
- **Q** Which AML stage best hides illicit origin? **Ans** Layering.
- **Q** Which obligation mandates identity verification? **Ans** KYC.
- **Q** Which red flag suggests threshold evasion? **Ans** just-below-threshold amounts / structuring.

- **Q** Which graph property finds influencers? **Ans** Centrality.
- **Q** Which graph approach reveals hidden KYC links? **Ans** Graph analytics.
- **Q** Which DBSCAN parameter measures neighborhood density range? **Ans** eps radius.
- **Q** Reduce false positives by improving what? **Ans** Precision.
- **Q** RAG versus MCP in one line? **Ans** document grounding vs live tool/system connection.
- **Q** Why use Ollama? **Ans** private local model running with lower cost and better control.